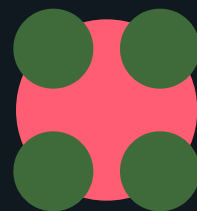


FALL 2026 / CURRENT PLAYABLE SPECIFICATION

BLOOM



Game Design Document

A browser-based top-down multiplayer survival shooter set in a procedural garden. Explore a vast world, arm yourself, grow a healing flower, and survive the closing storm.

31,000 x 31,000

BASE WORLD SIZE

30 Hz

AUTHORITATIVE SIMULATION

1 - 50 HP

FLOWER GROWTH

FORMAT

Online multiplayer / private rooms / host-started matches

GENRE AND PLATFORM

2D top-down battle royale / desktop web browser

SECTION 01

Core loop and controls

Players enter unarmed inside a populated central garden. They collect a rifle, ammunition, and one dormant seed; choose when to fight or reposition; and stay inside randomized nested safe zones until one survivor remains.

01 EXPLORE

Read the tiled world, move through connected corridors, and use walls and crate stacks as hard cover.

02 COLLECT

Auto-collect a rifle, spare ammunition, and one seed. Starting-area resources keep the first decision close.

03 ENGAGE

Aim with the mouse, hold fire for automatic shots, use cover, and track bright projectile trails and impacts.

04 SURVIVE

Reposition as the storm contracts. Consume the carried flower when its current healing value is worth the trade.

CONTROLS

WASD

Move

MOUSE

Aim

HOLD CLICK

Fire

R

Reload

E

Heal

SECTION 02

World, scale, and storm

The map is twenty times the original width and height while players, movement, collision bodies, and 64-unit floor tiles keep their gameplay scale. This creates real travel distance without zooming the camera.

31,000

BASE WIDTH AND HEIGHT

169,000

64-PLAYER WIDTH AND HEIGHT

20x

LINEAR WORLD SCALE

POPULATED START

Spawns remain in a dense central garden with protected radii, connected cross-corridors, rifles, ammunition, seeds, and readable cover.

EXPANDED OUTSKIRTS

Additional cover clusters and loot populate the larger world. Counts remain bounded by player population so simulation and network costs stay controlled.

STORM PACING

Each contraction lasts at least 60 seconds. On large circles the server lengthens the phase so the fastest moving boundary remains below 65 percent of player running speed. The storm then holds for 10 seconds before selecting another nested target.

- Outside the safe zone: 1 HP damage per second.
- The server publishes the actual phase duration for an accurate client countdown.
- Later small circles return to the 60-second minimum.

SECTION 03

Combat and bloom

AUTOMATIC RIFLE

30-round magazine. 5 damage per hit. 120 ms fire interval. 1.35 second reload. Spread is small and tunable. Ammunition pickups add one magazine.

PROJECTILES

Bullets travel at 920 units per second until their first wall, player, or map-boundary collision. There is no arbitrary short lifetime.

COLLISION AND FEEDBACK

- Continuous swept collision prevents fast bullets from skipping thin cover.
- Relative motion catches a player crossing a bullet path between simulation ticks.
- Muzzle obstruction stops a protruding barrel from firing or lighting through a wall.
- The actual projectile sprite, 90 ms trail, muzzle flash, recoil, impact sparks, and warm light communicate every shot.

FLOWER GROWTH



SEED

1 - 16 HP

Weak now, strong later.



BUDDING

17 - 33 HP

A useful emergency heal.



BLOOMED

34 - 50 HP

Maximum mechanical payoff.

SECTION 04

Art direction and readability

Strictly top-down, crisp, modular sprite art over a dark slate tiled world. The presentation favors immediate combat readability, with selective lighting and effects providing direction and impact.

CHARACTER RIG

Separate body, backpack, head, and arms. The backpack sits behind the shoulders. Arms attach at shoulder pivots and use distinct resting and rifle-grip poses.

WORLD ART

Subdued 64-unit floor tiles, lighter central corridors, blue walls with gold trim, and crate stacks drawn exactly inside collision bounds.

LIGHT

Cool ambient tone, soft player halos, aimed rifle beams, pickup glows, and warm muzzle and impact illumination. Cover casts hard directional shadows.

PROJECTILE READ

A visible sprite leads a gold-white trail. Trail and light terminate at the authoritative collision point, then fade quickly without implying continued travel.

PALETTE

Night slate

Wall blue

Gold trim

Bloom pink

Warm white

SECTION 05

Technical scope and risks

CLIENT

HTML, CSS, JavaScript, and Canvas hosted on Vercel. Local movement prediction reconciles with the server; remote players interpolate between snapshots.

NETWORK RISK

Shot and impact timestamps preserve very short projectiles between 15 Hz snapshots. Reconnect tokens and multi-node room partitioning remain future work.

SERVER

Node.js WebSocket service on one AWS EC2 instance. Match state, collision, damage, loot, storm, elimination, and spectating are authoritative.

WORLD RISK

The expanded world can lengthen travel and matches. Keep spawn resources dense, measure engagement time, and tune outer cover and loot from playtests.

CAPACITY

A 64-player room limit is an implementation safety rail. World, cover, and loot scale by population; practical capacity still requires measured playtests.

PERFORMANCE RISK

Visible world tiles and cover are culled. Lighting uses a half-resolution viewport buffer and capped nearby lights. Collision still scales linearly with cover.

CURRENT DEPLOYMENT

Client: Vercel / Server: AWS EC2 behind HTTPS and WSS / Source: GitHub master

PLAYTEST QUESTION

Does the expanded world create meaningful pursuit and exploration while the storm and dense opening garden still produce timely, readable combat?